

GST-Ready Double-Entry Accounting Ledger

A clean-room rebuild of the Bigcapital accounting core: balanced journals, reversible invoices, exact money.

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backend tests

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frontend tests

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Killer Tests

What the audit found, and what we changed

REFERENCE (BIGCAPITAL) AUDIT

Balance updates not serialized with other writes

No debit = credit check before saving journals

Floating-point money

No duplicate-payment detection

OUR REBUILD

→ BEGIN IMMEDIATE write lock + UNIQUE document numbers

→ validateJournal rejects with HTTP 400 before any write

→ Integer cents everywhere; fractions rejected

→ Non-blocking duplicate / unusual amount alert

One ledger engine, five tiers

Presentation

React 18 + Vite + TypeScript

API gateway

Express 4 · JWT · role guards

Business services

Invoices · Payments · Bills · Reports

Ledger engine

Validation · balances · reversals

Database

SQLite · WAL · BEGIN IMMEDIATE

INVARIANTS ENFORCED

$\Sigma \text{ debits} = \Sigma \text{ credits}$

checked before every write

Single-sided ledger rows

CHECK constraint in schema.sql

Atomic writes

journal + balances in one transaction

Reversal by inversion

original rows inverted, invoice kept as Voided

Trial balance from ledger

aggregated from accounts_transactions

The three Killer Tests all pass

KT-1

Journal balance

Journal with debit 10000 and credit 8000 sent to POST /api/manual-journals.

RESULT

HTTP 400. Zero rows written. Balances untouched.

KT-2

Invoice reversal

\$450 invoice created, delivered, then voided with DELETE /api/sale-invoices/:id.

RESULT

4 ledger rows (2 + 2 inverse). Voided, row kept. Back to \$0.

KT-3

Randomized operations

20 seeded random operations (seed 987654321): invoices, payments, bills, voids.

RESULT

Trial balance: debits = credits, difference \$0.00.

Two improvements, one demo-ready product

Improvement 1 • Payment alert

Flags the same customer paying the same amount within 10 minutes, or an amount 5x or more above their average (3+ prior payments). Advisory only; the payment still saves.

Improvement 2 • Write safety

BEGIN IMMEDIATE takes the SQLite write lock before any accounting write. True UNIQUE constraints on invoice, bill, payment and journal numbers, and account codes.

THE APP

- Dashboard with LEDGER BALANCED badge
- Invoices: Draft → Delivered → Voided
- Payments, bills, manual journals
- Trial balance, balance sheet, P&L
- Admin: users and KT-3 harness

OUT OF SCOPE

Multi-currency, inventory/COGS, bank feeds, payment gateways.